

Exercise 1 — Bubble Sort / Selection Sort (General)

Using a static `int [ ]` array, sort the array in ascending order using both the bubble sort and selection sort algorithms.

Exercise 2 — Selection Sort / Bubble Sort

Implement the Selection Sort or Bubble Sort algorithm to sort the list of integers [34, 12, 25, 9, 78, 31]. Print the sorted list.

Exercise 3 — Selection Sort / Bubble Sort with Step Tracking

Write a program to implement the Selection/Bubble Sort algorithm and sort the list of integers [45, 11, 32, 5, 27, 19] in ascending order, then print the sorted array.

Optional bonus: Count and print the number of comparisons and swaps performed.

Exercise 4 — Selection Sort / Bubble Sort

Write a program to implement the Selection/Bubble Sort algorithm and sort the list of integers [29, 10, 14, 37, 13]. Print the sorted list.

Exercise 5 — Selection Sort (with Intermediate Steps)

Implement the Selection Sort algorithm to sort an array of integers in ascending order.

Input: An array of integers.

Output: - The original array before sorting. - Intermediate steps showing the array after each pass of Selection Sort. - The final sorted array in ascending order.

Exercise 6 — Insertion Sort / Merge Sort

Implement Insertion Sort or Merge Sort to sort the list of integers [45, 11, 32, 5, 27, 19] in ascending order, and print the sorted array.

Optional bonus: Count and print the number of comparisons and swaps (if any).

Exercise 7 — Linear Search

Implement the Linear Search algorithm to search for a number in the list of integers [45, 23, 67, 12, 34, 89, 50]. Take a number as input from the keyboard and return its index if found, otherwise return -1.

Exercise 8 — Binary Search

Implement Binary Search to find a number in the sorted array [2, 5, 8, 12, 16, 23, 38, 56, 72, 91].

- Input a number from the keyboard.
- Return the index (0-based) if found, otherwise return -1.

Note: Do not use a built-in library search function.

Exercise 9 — Binary Search

Implement the Binary Search algorithm to search for a number in the sorted array [3, 6, 9, 12, 15, 18, 21, 24]. Take a number as input from the keyboard and return its index if found, otherwise return -1.

Exercise 10 — Longest Palindromic Substring

Write a program to find the longest palindromic substring in a given string.

Input: A string of characters.

Output: - The longest palindromic substring in the given string. - If no palindromic substring exists (of length greater than 1), print “No palindromic substring found.”

Example: Input: “abacdfgdcabba”